

Data Structure Practice Questions - Answers

Q1

A data structure is a way of organizing and storing data so it can be accessed and modified efficiently.

Examples: Array (student marks), Stack (browser back button), Queue (people waiting in a line).

Q2

Linear: elements arranged sequentially; examples Array, Linked List, Stack, Queue.

Non-linear: hierarchical/network; examples Tree, Graph.

Q3

Array stores same-type elements in contiguous memory. Diagram:

Index: 0 1 2 3

Memory: [10][20][30][40]

Q4

Algorithm: 1. Input array, pos, item. 2. Shift elements right from end to pos. 3. Insert item. 4. Increase size.

C Code:

```
for(i=n-1; i>=pos; i--) a[i+1]=a[i]; a[pos]=item; n++;
```

Q5

Stack follows LIFO (Last In First Out). Push adds at TOP, Pop removes from TOP.

Diagram: TOP->[30]->[20]->[10]

Q6

Stack: LIFO, one end (TOP). Queue: FIFO, insertion at rear, deletion at front.

Q7

Enqueue: rear++; q[rear]=item. Dequeue: item=q[front]; front++;

Q8

Singly Linked List: each node contains Data and Next pointer.

Diagram: [10|*] -> [20|*] -> [30|NULL]